

8.334 Midterm Review Session

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The Ising Ferromagnet

Consider the ferromagnetic Ising model on a d -dimensional hypercubic lattice, with Hamiltonian:

$$\mathcal{H} = - \sum_{\langle i,j \rangle} J s_i s_j - \sum_i H s_i \quad (1)$$

where $J > 0$, $s_i \in \{-1, 1\}$, and $i \in \mathbb{Z}^d$.

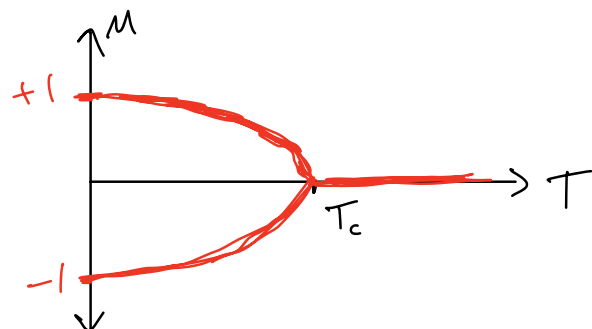
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We define the magnetization order parameter:

$$M = \frac{1}{N} \sum_i s_i$$

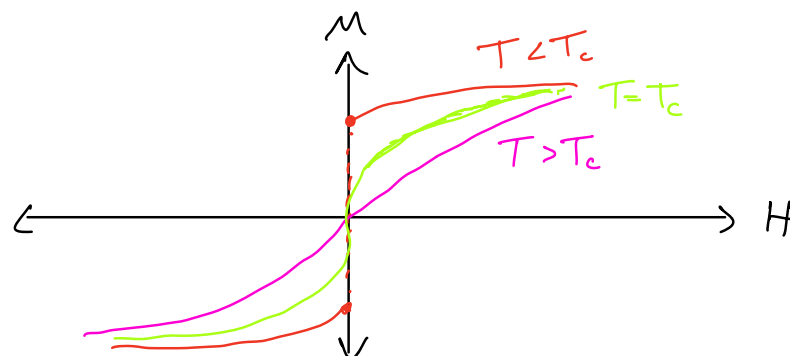
This is found to display an order-disorder transition:

For $h = 0$



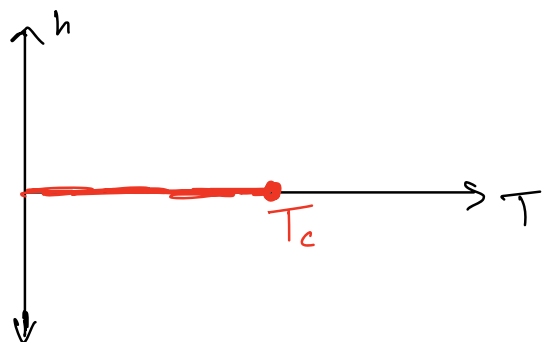
As T is lowered below T_c , the $M \rightarrow -M$ symmetry is spontaneously broken.

For $h \neq 0$



When h changes sign, M changes sign smoothly for $T > T_c$ but discontinuously for $T < T_c$. At $T \rightarrow T_c$, M is continuous but $\partial_H M|_{h=0} \rightarrow \infty$.

We can summarize with a phase diagram:



M varies analytically along any path that doesn't cross the singular coexistence line or the critical point ($h=0, T=T_c$).

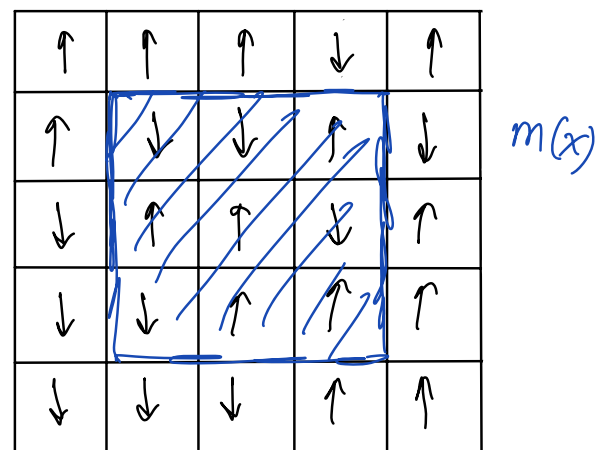
Landau - Ginzburg Theory

To study the singular behavior, we postulate the existence of a mesoscopic coarse-grained field, representing some local average of the magnetization

over many lattice sites. We call this $m(x)$:

$$m: \mathbb{R}^d \rightarrow \mathbb{R}$$

(notice that we have passed from $\mathbb{Z}^d$ to $\mathbb{R}^d$, and let m take values outside $[-1, 1]$. In these notes, I restrict to scalar m , i.e. $n=1$).



In principle, the Hamiltonian for m may be obtained by an appropriate coarse graining procedure from the microscopic Hamiltonian (1). This is hard in general. We instead include all terms allowed by symmetry:

$$\beta \mathcal{H}[m] = \int d^d x \left[\frac{t}{2} m^2 + u m^4 + \frac{K}{2} (\nabla m)^2 + \dots - h m(x) \right]$$

$|m|$ wants to be 0 or >0 depending on sign of t

m is banded below and above ($u > 0$)

spins want to locally align ($K > 0$)

Alignment with external field

The phenomenological parameters ($t, u, K, h, \dots$) are unspecified functions of the three microscopic parameters (J, H, T).

The probability of a field configuration $m(x)$ is given by the Boltzmann weight:

$$P[m(x)] \propto e^{-\beta \mathcal{H}[m]}$$

Partition function: $\mathcal{Z} = \int Dm(x) \exp[-\beta \mathcal{H}[m]]$

Free energy: $\beta F \equiv -\log \mathcal{Z}$

Mean field theory

Suppose the magnet always adopts its most probable configuration. Since $K > 0$, the most probable state must be uniform $\nabla m = 0$. Thus:

$$\mathcal{Z}_{\text{MF}} = \int dm \exp(-V \psi(m)), \quad \psi(m) = \frac{t}{2} m^2 + u m^4 - h m$$

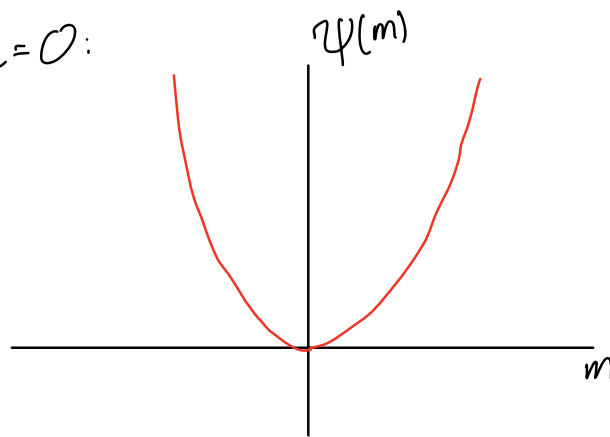
For $V \rightarrow \infty$, the leading saddle-point result is:

$$\beta F_{\text{MF}} = -\log \mathcal{Z}_{\text{MF}} = V \min_m \{ \psi(m) \},$$

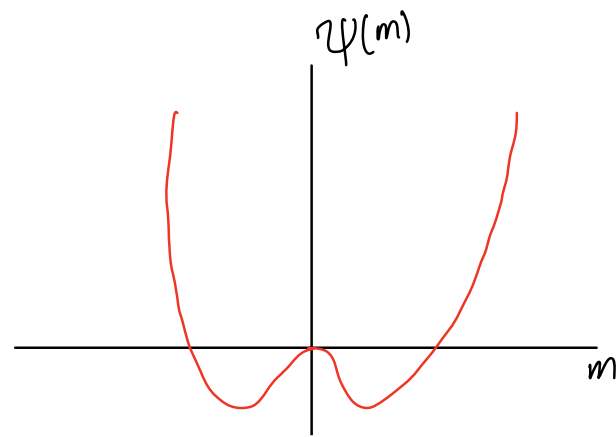
Call the magnetization $\bar{m} \equiv \arg \min_m \psi(m)$, which is the soln to $t \bar{m} + 4u \bar{m}^3 = h$

$\psi(m)$ changes shape at $t=0$:

We thus identify
 $t \propto T - T_c$.



$t > 0, h = 0$



$t < 0, h = 0$

Mean Field Critical Exponents:

We can compute the magnetization:

$$\bar{m}(h=0) = \begin{cases} 0 & t > 0 \\ \sqrt{\frac{-t}{4u}} & t < 0 \end{cases}$$

We thus obtain the critical exponent $\beta = 1/2$, defined via $m \propto |T - T_c|^\beta$.

Can also verify

$$\bar{m}(t=0) = \left(\frac{h}{4u}\right)^{1/3} \Rightarrow \delta = 3, \text{ where } \bar{m} \propto |h|^{1/\delta}$$

The susceptibility diverges as:

$$\chi = \left. \frac{\partial \bar{m}}{\partial h} \right|_{h=0} \propto 1/t \Rightarrow \gamma = 1, \text{ where } \chi \propto |T - T_c|^{-\gamma}$$

Heat capacity is discontinuous:

$$\beta F = V \times \begin{cases} 0 & \text{for } t > 0 \\ -t^2/16u & \text{for } t < 0 \end{cases}$$

$$\Rightarrow C(h=0) = -T \frac{\partial^2 F}{\partial T^2} \propto \begin{cases} 0 & \text{for } t > 0 \\ 1/8u & \text{for } t < 0. \end{cases}$$

Fluctuation Corrections to the Saddle point

Are the saddle point critical exponents stable to small fluctuations about the uniform profile? We will see that the answer is no.

Let us write

$$m(\vec{x}) = \bar{m} + \phi(\vec{x})$$

and expand the Hamiltonian to quadratic order in ϕ :

$$\beta \mathcal{H} = V \left(\frac{t}{2} \bar{m}^2 + u \bar{m}^4 \right) + \int d^d \vec{x} \left[\frac{\kappa}{2} (\nabla \phi)^2 + \underbrace{\frac{t + 12u \bar{m}^2}{2}}_{\kappa/2\Xi^2} \phi^2 \right]$$

To solve this, work in Fourier Space:

$$\phi(\vec{q}) \equiv \int d^d \vec{x} e^{-i\vec{q} \cdot \vec{x}} \phi(x) \Rightarrow \phi(\vec{x}) = \int \frac{d^d \vec{q}}{(2\pi)^d} e^{+i\vec{q} \cdot \vec{x}} \phi(\vec{q})$$

$$\kappa/2\Xi^2, \text{ where } \Xi^2 = \begin{cases} 1/\kappa t & t > 0 \\ 1/2\kappa|t| & t < 0 \end{cases}$$

where I have used the identity: $\int d^d \vec{q} e^{-i\vec{q} \cdot \vec{x}} = (2\pi)^d \delta(\vec{x})$.

Using this identity, it can be verified:

$$\int d^d \vec{x} \frac{\kappa}{2} [(\nabla \phi)^2 + \Xi^{-2} \phi^2] = \int \frac{d^d q}{(2\pi)^d} \frac{\kappa}{2} [q^2 + \Xi^{-2}] |\phi(q)|^2$$

a) Correlation function

The probability of a fluctuation configuration $\phi(q)$ is thus

$$\mathcal{P}(\{\phi(q)\}) \propto \exp \left[\int \frac{d^d q}{(2\pi)^d} \frac{\kappa}{2} [q^2 + \Xi^{-2}] |\phi(q)|^2 \right]$$

i.e. $\phi(q)$ is a Gaussian random field with the (diagonal) kernel: (Sec 3.5)

$$\mathcal{K}(q, q') = \frac{\delta(q+q')}{(2\pi)^d} \kappa(q^2 + \Xi^{-2})$$

Inverse kernel satisfies $\int d^d q' \mathcal{K}(q, q') \mathcal{K}^{-1}(q', q'') = \delta(q - q'')$

$$\frac{1}{(2\pi)^d} \mathcal{K}^{-1}(-q, q'') \kappa(q^2 + \Xi^{-2}) = \delta(q - q'')$$

$$\Rightarrow \mathcal{K}^{-1}(q, q'') = \frac{(2\pi)^d \delta(q + q'')}{\kappa(q^2 + \Xi^{-2})}$$

Thus:

$$\langle \phi(q) \phi(q') \rangle_c = \mathcal{K}^{-1}(q, q') = \frac{(2\pi)^d \delta(q + q')}{\kappa(q^2 + \Xi^{-2})}$$

In real space:

$$\langle \phi(x) \phi(x') \rangle = \frac{1}{\kappa} \int \frac{d^d \vec{q}}{(2\pi)^d} \frac{e^{-i\vec{q} \cdot (x - x')}}{q^2 + \Xi^{-2}}$$

It can be shown (Sec 3.2), by writing the above as the solution to a

screened Poisson equation, that the long-distance behavior follows:

$$\langle \phi(\vec{x}) \phi(\vec{x}') \rangle \sim \begin{cases} r^{1-\frac{d}{2}} e^{-r/\xi} & \text{for } t \neq 0 \ (\xi < \infty). \\ r^{2-d} & \text{for } t = 0 \ (\xi \rightarrow \infty). \end{cases}$$

where $r \equiv |\vec{x} - \vec{x}'|$

In experiments, however, one finds $\langle \phi(x) \phi(x') \rangle \sim r^{2-d-\gamma}$.

b) Free energy and heat capacity.

The partition function is:

$$Z = e^{-V(t\bar{m}^2/2 + u\bar{m}^4)} \int \mathcal{D}[\phi] \exp \left[- \int \frac{d^d q}{(2\pi)^d} \frac{K}{2} [q^2 + \xi^2] |\phi(q)|^2 \right]$$

$$\propto e^{-\beta F_{MF}} \times \frac{1}{\sqrt{\det \mathcal{K}(q, q')}}}$$

$$q^{d-1} \log(q^2)$$

$$\begin{aligned} \Rightarrow \beta F &= -\log Z = \beta F_{MF} + \frac{1}{2} \log \det \mathcal{K} \\ &= \beta F_{MF} + \frac{1}{2} \text{Tr} \log \mathcal{K} \\ &= \frac{t}{2} \bar{m}^2 + u \bar{m}^4 + \frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \log \lambda(q) \end{aligned}$$

where $\lambda(q)$ denotes the eigenvalues of $\mathcal{K}(q, q')$, so that

$$\beta F = \frac{t}{2} \bar{m}^2 + u \bar{m}^4 + \frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \log (K(q^2 + \xi^{-2}))$$

The heat capacity is obtained by noting that $\xi^{-2} \propto t$:

$$C = - \frac{\partial^2 \beta F}{\partial t^2} = \begin{cases} 0 & + \frac{1}{2} C_F \\ \frac{1}{8} u & + 2 C_F \end{cases}$$

where

$$C_F = \frac{1}{K^2} \int \frac{d^d \vec{q}}{(2\pi)^d} \frac{1}{(q^2 + \xi^{-2})^2} \propto \int_0^\Lambda dq \frac{q^{d-1}}{(q^2 + \xi^{-2})^2}$$

For large q , integrand goes as q^{d-4} . For $d > 4$, we have cutoff dependent constant: $C_F \propto \Lambda^{d-4}$ ($d > 4$)

For $d < 4$, the integral converges in the UV. Rescaling by $\tilde{q} = \xi q$ to make C_F dimensionless, we find

$$C_F \propto \xi^{4-d} \quad \text{for } d < 4.$$

Thus as $t \rightarrow 0$, the correction diverges as $C_F \propto t^{\frac{4-d}{2}} \Rightarrow \alpha = \frac{4-d}{2}$.

Since this correction dominates the baseline saddle point result, we can only conclude that saddle point + Gaussian fluctuations is unreliable in $d < 4$.

Direct Perturbation theory

Given that the uniform, saddle point solution + Gaussian fluctuations is unreliable in $d < 4$, we may ask whether the nonlinearity $\propto u m^4$ saves the day. To answer this, we will attempt to include $u m^4$ perturbatively, writing:

$$\beta \mathcal{H} = \beta \mathcal{H}_0 + U = \int d^d \vec{x} \left[t \frac{m^2}{2} + K (\nabla m)^2 \right] + u \int d^d \vec{x} m^4$$

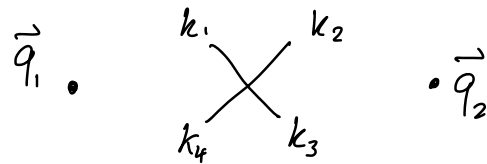
The expectation value of any observable is then given by:

$$\begin{aligned} \langle \mathcal{O} \rangle &= \frac{\int \mathcal{D}[\vec{m}] \mathcal{O} e^{-\beta \mathcal{H}_0 - U}}{\int \mathcal{D}[\vec{m}] e^{-\beta \mathcal{H}_0 - U}} = \frac{\int \mathcal{D}[\vec{m}] \mathcal{O} e^{-\beta \mathcal{H}_0} [1 - U + U^2/2 + \dots]}{\int \mathcal{D}[\vec{m}] e^{-\beta \mathcal{H}_0} [1 - U + U^2/2 + \dots]} \\ &= \frac{\langle \mathcal{O} \rangle_0 - \langle \mathcal{O} U \rangle_0 + \langle \mathcal{O} U^2 \rangle_0 / 2 - \dots}{1 - \langle U \rangle_0 + \langle U^2 \rangle_0 / 2 - \dots} \\ &\equiv \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \langle \mathcal{O} U^n \rangle_0^c \end{aligned}$$

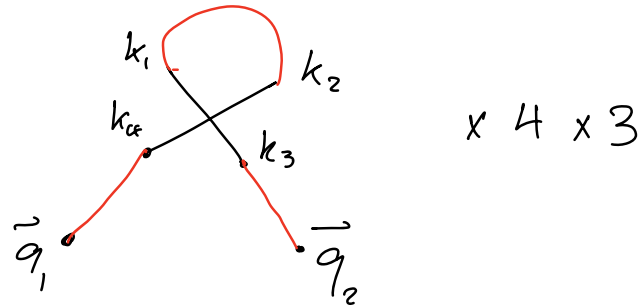
We will compute the two-point function $\langle m(q) m(q') \rangle$ to order $\mathcal{O}(u)$.

The free propagator is $\langle m(\vec{q}) m(\vec{q}') \rangle_0 = \frac{(2\pi)^d \delta(\vec{q} + \vec{q}')}{t + K q^2}$.

1) Draw 2 external legs $\vec{q}$ and $\vec{q}'$. Draw one vertex.



2) Make all pairwise connections between external and internal legs, discarding disconnected diagrams. In this case, there is only one diagram at $\mathcal{O}(u)$



where the numerical factor of 12 counts the # of topologically equivalent pairings

3) Replace every connection with $\vec{q} \rightarrow \vec{q}' = \frac{(2\pi)^d \delta(\vec{q} + \vec{q}')}{t + K q_1^2}$

every vertex with $\begin{matrix} k_1 & k_2 \\ & \times \\ k_3 & k_4 \end{matrix} = u (2\pi)^d \delta(k_1 + k_2 + k_3 + k_4)$

4) Integrate over internal momenta.

All in all:

$$\begin{aligned}
 \langle m(q_1) m(q_2) \rangle &= \vec{q}_1 \rightarrow \vec{q}_2 - 12 \times \begin{matrix} & & & \\ & \times & & \\ & & & \\ q_1 & & & q_2 \end{matrix} \\
 &= \frac{(2\pi)^d \delta(q_1 + q_2)}{t + K q_1^2} - 12 u \int \frac{d^d k_1 d^d k_2 d^d k_3 d^d k_4}{(2\pi)^{4d}} \frac{(2\pi)^d \delta(q_1 + k_1)}{t + K q_1^2} \cdot \frac{(2\pi)^d \delta(q_2 + k_2)}{t + K q_2^2} \\
 &\quad \cdot \frac{(2\pi)^d \delta(k_3 + k_4)}{t + K k^2} \cdot (2\pi)^d \delta(k_1 + k_2 + k_3 + k_4) \\
 &= \frac{(2\pi)^d \delta(q_1 + q_2)}{t + K q_1^2} - 12 u \int \frac{d^d k}{(2\pi)^d} \frac{(2\pi)^d \delta(q_1 + q_2)}{(t + K q_1^2)^2 (t + K k^2)} \\
 &= \frac{(2\pi)^d \delta(q_1 + q_2)}{t + K q_1^2} \left[1 + \frac{12 u}{t + K q_1^2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{t + K k^2} + \mathcal{O}(u^2) \right]
 \end{aligned}$$

We can compute the susceptibility using

$$\chi = \int d^d x \langle m(x) m(0) \rangle = \lim_{q \rightarrow 0} \langle |m(q)|^2 \rangle$$

We find

$$\chi^{-1}(t) = (t - t_c) \left[1 - \frac{12u}{K^2} c \left(\frac{K}{t - t_c} \right)^{2-d/2} + \mathcal{O}(u^2) \right]$$

One again: At the critical point, correction dominates the bare value.

$\Rightarrow$ Perturbation theory unreliable.

Scaling Theory

We require an alternate path towards critical exponents besides that provided by standard perturbation theory.

Can check from mean field theory that the free energy density has the following homogeneous form near criticality:

$$F(t, h) = |t|^2 g_F(h/t^{3/2})$$

The homogeneity assumption states that in the full system, the critical free energy maintains this form:

$$F(t, h) = |t|^{2-\alpha} g_F(h/t^\Delta)$$

By taking derivatives, all critical exponents may be obtained

At criticality, the correlation function goes as ($\xi \rightarrow \infty$)

$$G(x) = \langle m(x) m(0) \rangle \sim 1/|x|^{d-2+\eta}$$

This implies a dilation symmetry

$$G(\lambda x) = \lambda^P G(x)$$

The critical system is thus described by a scale-invariant theory. Such theories correspond to specific points in the infinite dimensional parameter space of the Landau-Ginzburg Hamiltonian $(t, h, K, u, L, \dots)$.

How do we find such points? In general (at or away from criticality), we may rescale space as $x_{\text{new}} = x_{\text{old}}/b$ and coarse grain the field. The new

theory is then described by a Hamiltonian with the same symmetries but different parameters

$$t, h, K, u \xrightarrow{x \rightarrow bx} t', h', K', u' \dots$$

The program of RG is to apply this procedure iteratively, looking for fixed points of this map. Such fixed points correspond to scale invariant theories where $\xi = 0$ or $\xi = \infty$.

To obtain critical exponents, we note that coarse graining should leave the free energy unchanged:

$$V f(t, h) = V' f(t', h')$$

Writing $t' = b^{y_t} t$, $h' = b^{y_h} h$, we find

$$f(t, h) = b^{-d} f(b^{y_t} t, b^{y_h} h)$$

Choosing $b = t^{-1/y_t}$, we get the homogeneous form:

$$f(t, h) = t^{d/y_t} g_f(h/t^{y_h/y_t}) \Rightarrow 2 - \alpha = d/y_t, \Delta = y_h/y_t$$

"anomalous dimensions".
Why $t' = b^{y_t} t$?
Because rescaling by b then b' is the same as rescaling by bb' .

Perturbative Renormalization Group

I will now briefly review Momentum-Shell RG. There are three steps:

1) Coarse grain: integrate out modes with $\Lambda/b < q < \Lambda$

2) Rescale: set $q = b^{-1} q'$ to restore cutoff to Λ

3) Renormalize: set $\tilde{m} = Z \bar{m}'$, fixing Z by demanding $K' = K$.

1) Coarse grain:

$$m(q) \equiv \begin{cases} \tilde{m}(q) & \text{for } q \in (0, \Lambda/b) \\ \sigma(q) & \text{for } q \in (\Lambda/b, \Lambda) \end{cases}$$

$$\begin{aligned} Z &= \int D\tilde{m}(q) D\sigma(q) \exp \left\{ - \int_0^\Lambda \frac{d^d q}{(2\pi)^d} \left(\frac{t + Kq^2}{2} \right) (|\tilde{m}(q)|^2 + |\sigma(q)|^2) - \mathcal{U}[\tilde{m}(q), \sigma(q)] \right\} \\ &= \int D\tilde{m} \exp \left[- \int_0^\Lambda \frac{d^d q}{(2\pi)^d} \left(\frac{t + Kq^2}{2} \right) |\tilde{m}(q)|^2 \right] \int D\sigma(q) e^{- \int_0^\Lambda \frac{d^d q}{(2\pi)^d} \left(\frac{t + Kq^2}{2} \right) |\sigma(q)|^2} e^{-u} \end{aligned}$$

$$\propto \int D\tilde{m} \exp^{-\int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \left(\frac{t + K q^2}{2} \right) |\tilde{m}|^2} \langle e^{-U[\tilde{m}, \sigma]} \rangle_\sigma$$

$$\equiv \int D\tilde{m} e^{-\beta \tilde{H}(\tilde{m})}$$

The coarse grained Hamiltonian is thus:

$$\beta \tilde{H} = \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \left(\frac{t + K q^2}{2} \right) |\tilde{m}(q)|^2 - \log \langle e^{-U[\tilde{m}, \sigma]} \rangle_\sigma$$

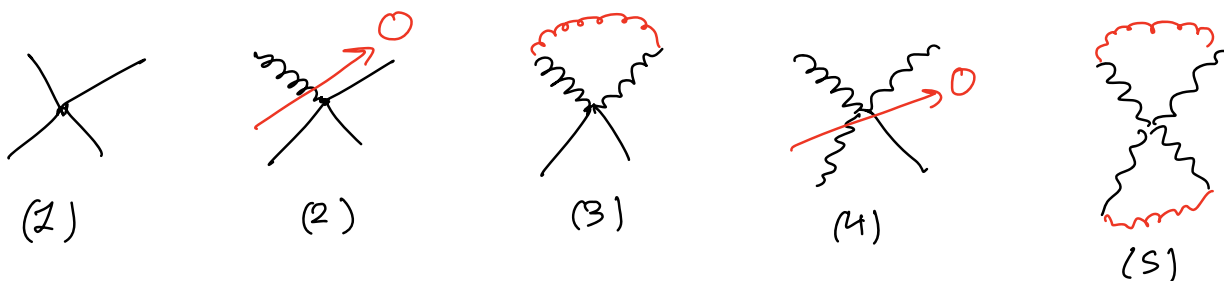
Final term can be computed diagrammatically:

$$\log \langle e^{-U} \rangle_\sigma = -\langle U \rangle_\sigma + \frac{1}{2} (\langle U^2 \rangle_\sigma - \langle U \rangle_\sigma^2) + \dots$$

First order:

$$\langle U \rangle_\sigma = u \int \prod_{i=1}^4 \frac{d^d q_i}{(2\pi)^d} (2\pi)^d \delta(q_1 + q_2 + q_3 + q_4) \left\langle \prod_{i=1}^4 (m(q_i) + \sigma(q_i)) \right\rangle_\sigma$$

Feynman rules: No "external legs," but we distinguish terms like $\tilde{m}$ and σ using a squiggly line for the latter. Possible diagrams



1) Just gives $U[\tilde{m}] = u \int d^d x m^4$

2, 4) Zero (odd # of σ)

5) Just a constant (vacuum bubble)

3) Only nontrivial term, renormalizes m .

$$4 \times 3 \cdot (-u) \cdot \int \frac{d^d q_1 \dots d^d q_4}{(2\pi)^{4d}} (2\pi)^d \delta(q_1 + \dots + q_4) \tilde{m}(q_1) \tilde{m}(q_2) \frac{(2\pi)^d \delta(q_3 + q_4)}{t + K q_3^2}$$

where $q_1, q_2 \in (0, \Lambda/b)$, $q_3, q_4 \in (\Lambda/b, \Lambda)$

$$= -12u \int_0^{\Lambda/b} \frac{d^d q_1 d^d q_2}{(2\pi)^{2d}} (2\pi)^d \delta(q_1 + q_2) \tilde{m}(q_1) \tilde{m}(-q_1) \int_{\Lambda/b}^{\Lambda} \frac{d^d q_3}{(2\pi)^d} \frac{1}{t + K q_3^2}$$

$$= -12u \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} |\tilde{m}(q)|^2 \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + K k^2}$$

Thus

$$\beta \tilde{H} = \int_0^{\Lambda/b} \frac{d^d q}{(2\pi)^d} \left(\frac{\tilde{t} + K q^2}{2} \right) |\tilde{m}(q)|^2 + u \int_{q < \Lambda/b} m^4$$

The mass is then shifted as:

$$\tilde{t} = t + 12u \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + K k^2}$$

Other terms: $\tilde{K} = K$, $\tilde{u} = u$.

2) Rescale: set $q = b^{-1} q'$, so that $\int_0^{\Lambda/b} d^d q \rightarrow \int_0^{\Lambda} d^d q' b^{-d}$

3) Renormalize: $\tilde{m} = Z m'$, so that

$$\beta H' = \int_0^{\Lambda} \frac{d^d q'}{(2\pi)^d} b^{-d} \left(\frac{\tilde{t} + K b^{-2} q'^2}{2} \right) Z^2 |m'(q)|^2 + u Z^4 b^{-3d} \int_0^{\Lambda} \frac{d^d q'_1 d^d q'_2 d^d q'_3}{(2\pi)^{3d}} m'_1 m'_2 m'_3 m'(-q_1 - q_2 - q_3).$$

$$\text{Thus: } t' = b^{-d} Z^2 \tilde{t} \quad K' = b^{-d-2} Z^2 K \quad u' = b^{-3d} Z^4 u$$

Demanding $K' = K$: $Z = b^{1+d/2}$.

$$t'_b = b^2 \left[t + b u \int_{\Lambda/b}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{t + K k^2} \right]$$

$$u'_b = b^{4-d} u.$$

Now write $b = 1 + \delta l$ and expand, to get the RG Flow:

$$\frac{dt}{dl} = 2t + \frac{6u K \Lambda^d}{t + K \Lambda^2}$$

$$\frac{du}{dl} = (4-d)u - Au^2$$

Problem: For $d < 4$, u flows to infinity. We require the next order to find the proper FP.

Following section 5.6, we get:

$$\frac{dt}{dl} = 2t + \frac{6uK_d\Lambda^d}{t+K\Lambda^2} - \mathcal{O}(u^2)$$

$$\frac{du}{dl} = (4-d)u - \frac{36K_d\Lambda^2}{(t+K\Lambda^2)^2} u^2$$

To make the **Wilson Fisher** fixed point close to the Gaussian one (and thus perturbatively accessible), we need $\varepsilon \equiv 4-d$ to be small. We then have FP values:

$$u^* = \frac{K^2}{36K_d} \varepsilon$$

$$t^* = -\frac{1}{6} K \Lambda^2 \varepsilon + \mathcal{O}(\varepsilon^2)$$

Linearizing about this FP, we obtain anomalous dimensions as the eigenvalues of the linearized RG flow: for an operator p :

$$p' = b^y p \simeq (1+\Delta l)^y (p^* + \delta p)$$

$$(p^* + \delta p') = (1+y\Delta l)(p^* + \delta p)$$

$$\frac{\delta p' - \delta p}{\Delta l} = y \delta p$$

$$\frac{d}{dl} \delta p = y \delta p \Rightarrow y \text{ is eigenvalue for operator } p.$$